Overview of Algorithms

General Information

Suppose we have the molecular graph of a linear polyene molecule C_nH_{n+2}. The molecular graph of a linear polyene is a caterpillar tree. This tree contains a central path, which is a zigzag graph path of length 'n'. The vertices of this path represent carbon atoms in the linear polyene molecule. Therefore, we will denote the vertices of this path as C-vertices, and its edges as C–C edges.

One pendant vertex is attached to each internal vertex of the graph path, and two pendant vertices are attached to each terminal vertex. All these pendant vertices represent hydrogen atoms in the linear polyene molecule. Therefore, we will call these vertices H-vertices, and the edges connecting C-vertices to H-vertices — C–H edges.

If we number all vertices and edges of the graph path from left to right, we obtain the path: $v_1, e_1, v_2, e_2, v_3, e_3, ..., v_{p-1}, e_{p-1}, v_p$, where v_i and e_i are the C-vertices and edges with number i, $p = n - 2$ — the number of internal vertices of the graph, q — the number of internal edges, calculated as $q = p - 1 = n - 3$, and n is the order of the graph path.

According to the conventional model of molecular graphs of linear polyenes, all C–C edge lengths are equal, and any two adjacent C–C edges form an angle of 120 degrees. Any two adjacent C–C and C–H edges also form an angle of 120 degrees. Therefore, each internal vertex c_k of the graph path can be assigned a binary code that reflects the direction of the rotation angle of edge e_{k+1} relative to edge e_k.

The sequence of vertex codes forms the binary vertex code of the molecular graph. In the paper by Yu.L. Nosov and M.N. Nazarov, a coding system was adopted in which a left turn is encoded by zero, and a right turn is encoded by one.

Molecules of isomers and conformers of linear polyenes have an even number of carbon atoms. Therefore, in this package, the number of vertices 'n' in the graph path is also even. For molecular graphs with suppressed hydrogen atoms, 'n' is the order of that graph. The number of edges in the graph path is odd. Each internal edge of the graph path can be in either trans or cis configuration. Therefore, each edge can also be assigned a binary code. The sequence of edge codes forms the binary edge code of the molecular graph.

In the paper by Yu.L. Nosov and M.N. Nazarov, the following coding system was adopted for vertices and edges: • when encoding vertices, a left turn is coded as zero, and a right turn as one; • when encoding edges, the cis configuration is coded as one, and the trans configuration as zero.

Property E1 of Edge Codes

The same paper established a relationship between the edge code $Z(P_n)$ and the vertex code $B(P_n)$ of the graph. In the adopted vertex and edge coding system, each i-th element z_i of the edge code $Z(P_n)$ of graph P_n corresponds to two elements b_i and b_{i+1} of the vertex code $B(P_n)$, such that: $z_i = 0$ if $b_i \neq b_{i+1}$ and $z_i = 1$ if $b_i = b_{i+1}$

It follows that from the edge code of the molecular graph one can obtain its vertex code, and vice versa — from the vertex code one can recover the edge code.

All odd-numbered edges of the graph path correspond to double bonds C=C in the carbon-carbon path of the linear polyene molecule, while even-numbered edges correspond to single bonds C–C.

External edges are not used in the edge code. Therefore odd elements of the edge code correspond to single C–C bonds, and even elements correspond to double C=C bonds. Rigid double bonds determine the configuration of linear polyene molecules, while single bonds determine their conformation. Thus even elements of the edge code encode the configuration of the isomer called I-code, and odd elements of the edge code encode the conformation called C-code.

Property E2 of Edge Codes

In the adopted coding system (consistent with the properties of isomers and conformers): • for molecular graphs of trans-isomers: all elements of I-code and all elements of C-code equal 0; • for molecular graphs of other isomers: at least one element of I-code equals 1, and all elements of C-code equal 0; • for molecular graphs of conformers of trans-isomers: all elements of I-code equal 0, and at least one element of C-code equals 1; • for molecular graphs of conformers of other isomers: at least one element of I-code equals 1 and at least one element of C-code equals 1. It follows that the class of the molecular graph can be determined from its edge code. Taking into account the relationship between vertex and edge codes, the class of the molecular graph can also be determined from the vertex code.

It is more convenient to use the sums of the elements of these lists instead of the full I-code and C-code lists. Let us denote sum of I-code elements as 'sumiso', and sum of C-code elements as 'sumcon'.

Then, under the adopted coding system for the molecular graphs of: - trans-isomers: ' sumiso = 0' and 'sumcon = 0' - other isomers: $sumiso \neq 0$ and $sumcon = 0$ - conformers of trans-isomers: $sumiso = 0$ and $sumcon \neq 0$ - conformers of other isomers: $sumiso \neq 0$ and $sumcon \neq 0$

Property E3 of Edge Codes

In the paper by Yu.L. Nosov and M.N. Nazarov it was proved (assertion 3) that the edge code of the molecular graph of a linear polyene determines its symmetry group: - If the code $Z(P_n)$ of graph P_n is not symmetric, then the graph P_n has symmetry group C_S; - If the code $Z(P_n)$ of graph P_n is symmetric and the middle element of the code is 1, then the graph has symmetry group C_{2h}; - If the code $Z(P_n)$ of graph P_n is symmetric and the middle element of the code is 0, then the graph has symmetry group C_{2v}.

Property B1 of Vertex Codes

The vertex code of the molecular graph of a linear polyene determines the symmetry group of the graph:

```
1. If the vertex code $B(G)$ of graph $G$ is not
symmetric, then the graph has symmetry group $C_S$;

2. If $B(G) = Rev(B(G))$, then the graph has symmetry
group $C_{2h}$;

3. If $B(G) = Inv(Rev(B(G)))$, then the graph has
symmetry group $C_{2v}$.
```

Note: Here $Rev(B(G))$ and $Inv(B(G))$ denote reversal and inversion respectively, and $Inv(Rev(B(G)))$ is the composition of reversal and inversion (see Terminology section).

On Equivalence Classes of Vertex and Edge Codes

For edge codes of molecular graphs of linear polyenes the following holds:

For vertex codes of molecular graphs of linear polyenes the following holds:

Assertion 1. The original graph G constructed from some code $B(G)$, and the graphs G^I, G^R, G^{IR} constructed from codes B^I, B^R, B^{IR}, obtained by inversion $BI = Inv(B)$, reversal $B^R = Rev(B)$, and composition of inversion and reversal $B^I = Rev(Inv(B))$ are isomorphic, geometrically similar, and form one congruence class.

Assertion 2. The molecular graph G^R constructed from the code $Rev(Z(G)) (obtained by reversing the code $Z(G)$ of some graph G) is isomorphic and congruent to graph G and together with it forms one congruence class.